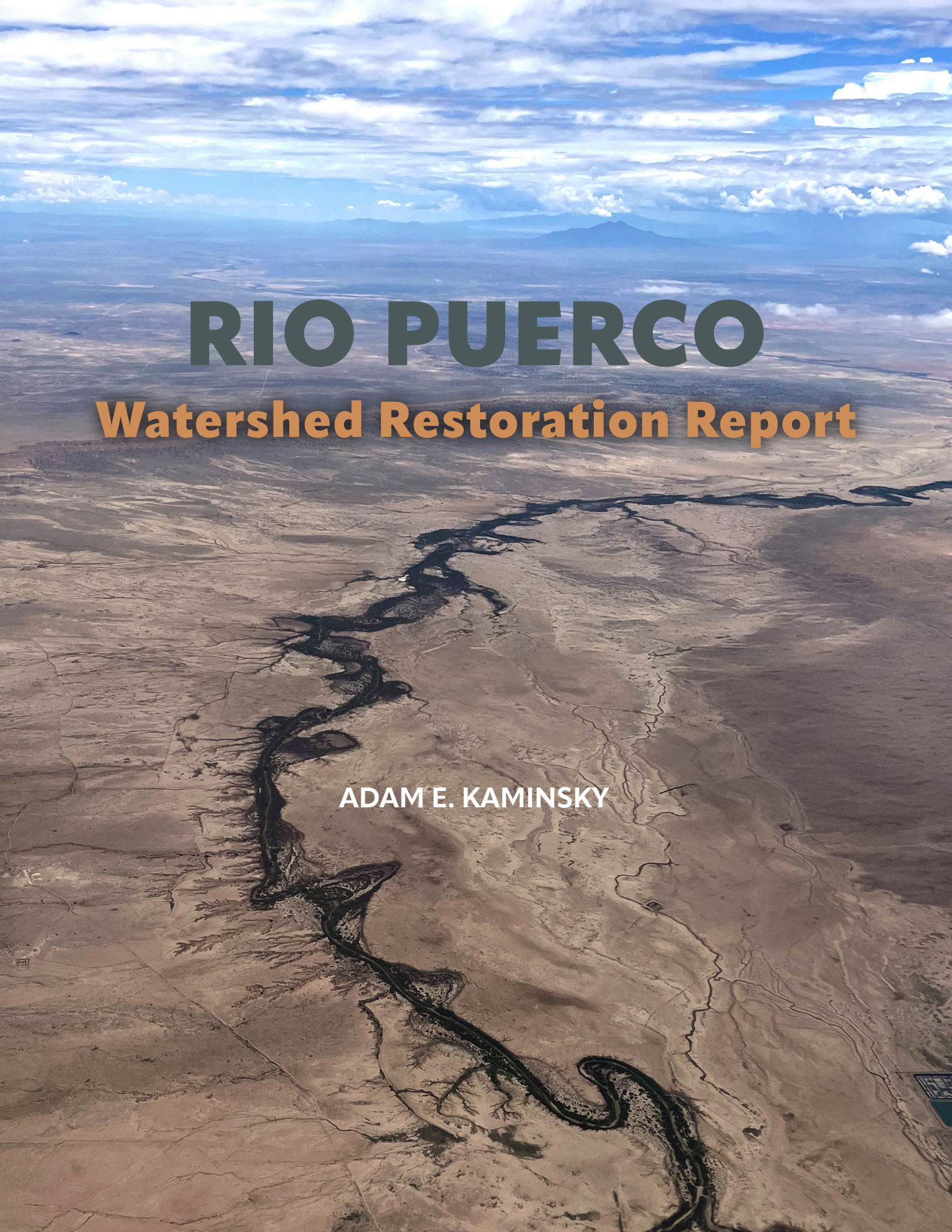

RIO PUERCO

Watershed Restoration Report

ADAM E. KAMINSKY

IN COLLABORATION WITH

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INTROD

UCTION

LOOKING EAST FROM HIDDEN MOUNTAIN ACROSS THE RIO PUERCO TO THE MANZANO MOUNTAINS

PROJECT DESCRIPTION

The purpose of this report is to support the diverse group of people and organizations who work within the Rio Puerco Watershed to improve the ecological, economic, and social health of the land.

Upper Rio Puerco in the foothills of the Nacimiento Mountains.

The report was compiled as part of a three-month internship with Quivira Coalition, for which the author researched publications, conducted informal interviews, volunteered with relevant organizations, and spent time exploring the watershed. Any views expressed are those of the author.

This report focuses on **three intervention recommendations** that further the goal of holistic restoration:

1 **Improve habitat and control erosion within the Rio Puerco and surrounding upland ecosystems.**

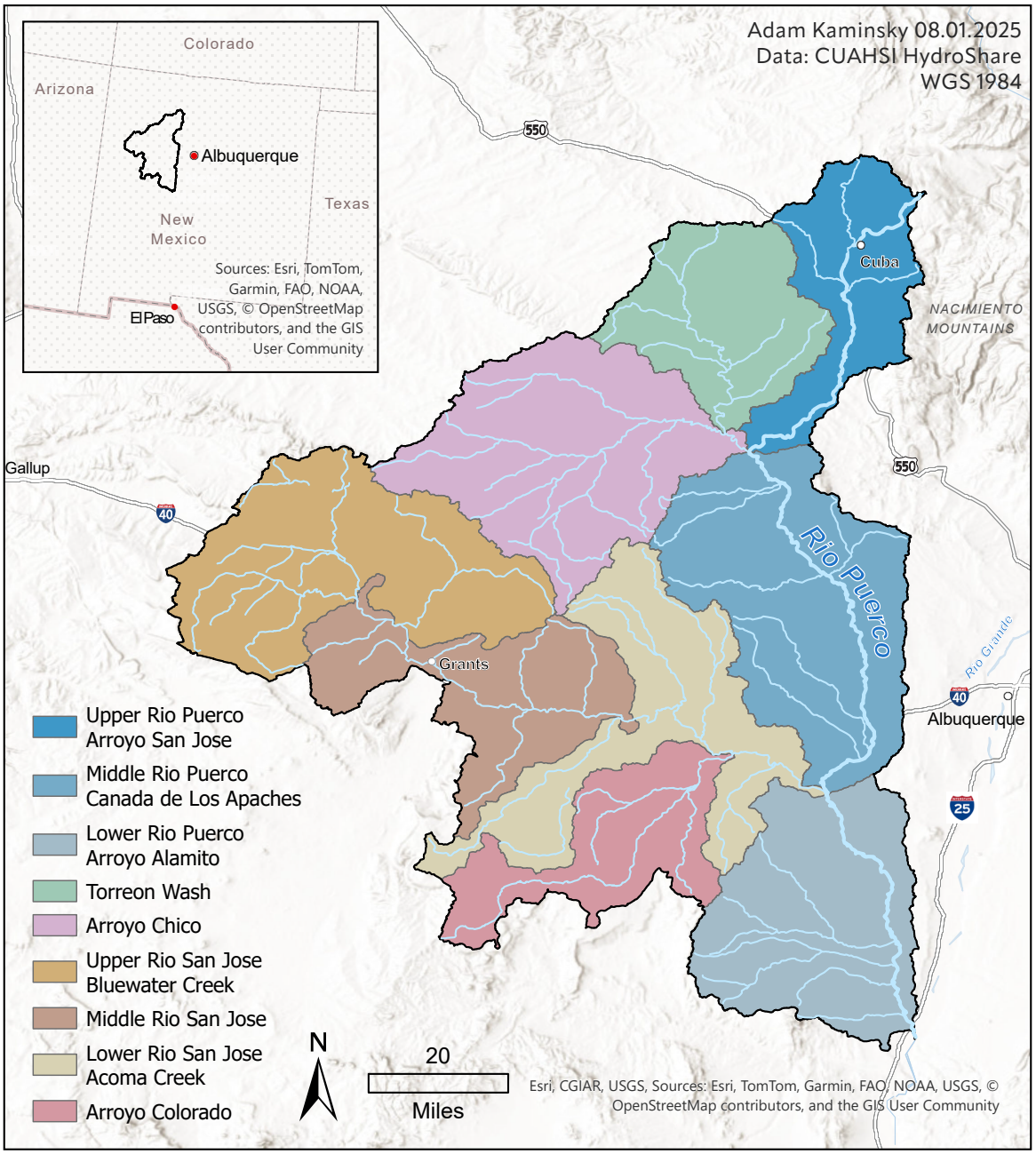
2 **Foster a community of practice that values sustainable and traditional land management and prioritizes erosion control.**

3 **Increase collaboration among relevant organizations across the fields of agriculture, conservation, and restoration.**

GEOGRAPHY OF THE REGION

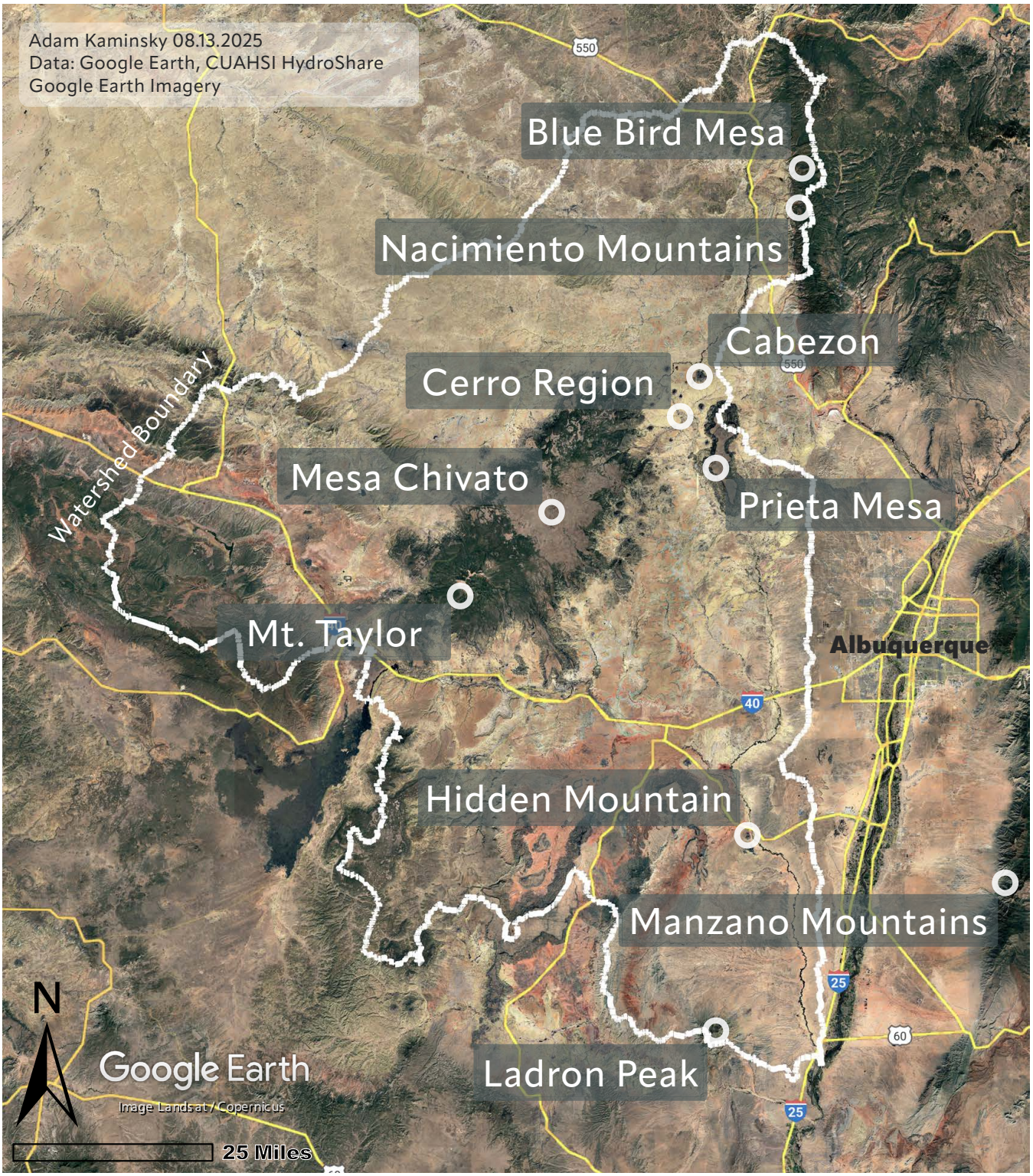
The Rio Puerco is situated in northwestern New Mexico, where it runs 150 miles south from the Nacimiento Mountains to the Rio Grande. The first 10 miles flows perennially, while the rest is ephemeral.

RIO PUERCO WATERSHED



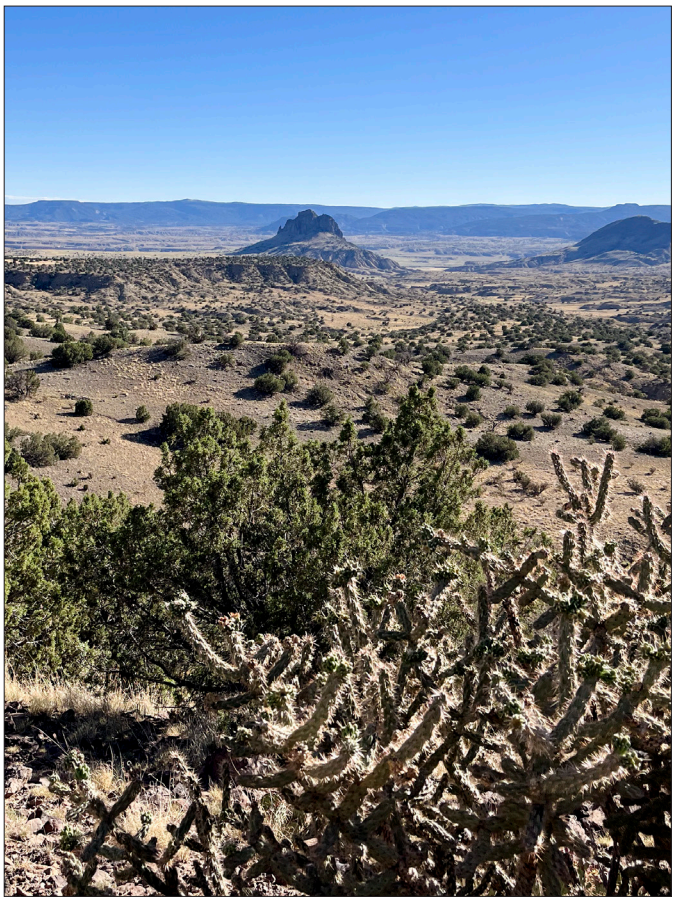
LANDMARKS

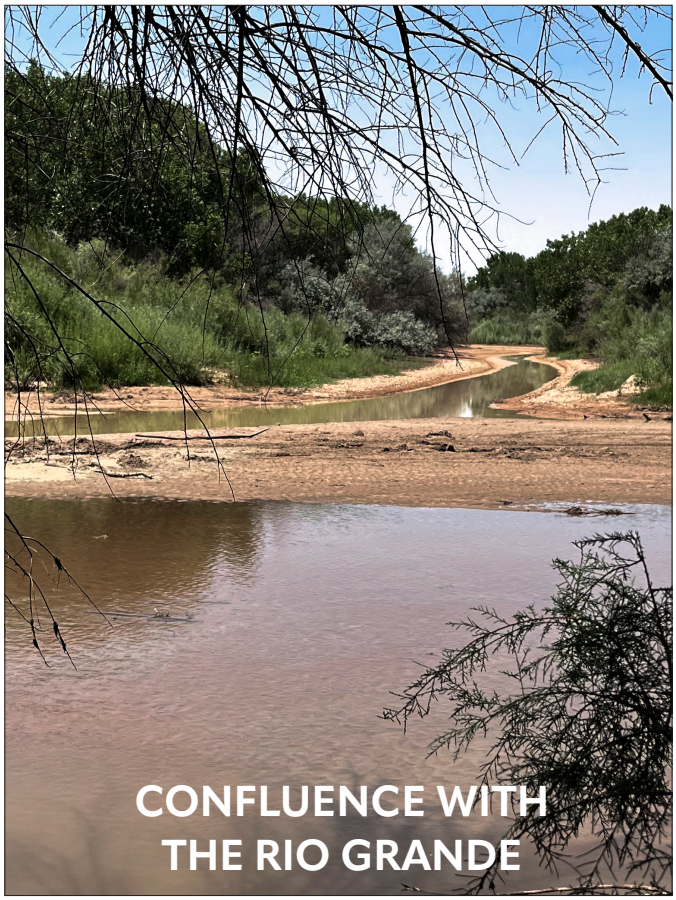
This map displays various landmarks noted in photographs throughout the report. Names reflect the common, colloquial naming of the region, and do not reference the numerous traditional Indigenous place names for these landmarks.



COLLOQUIAL LANDMARK NAMES IN THE RIO PUERCO WATERSHED

HEADWATERS REGION





CONFLUENCE WITH THE RIO GRANDE

LOWER WATERSHED

HIST

TORY

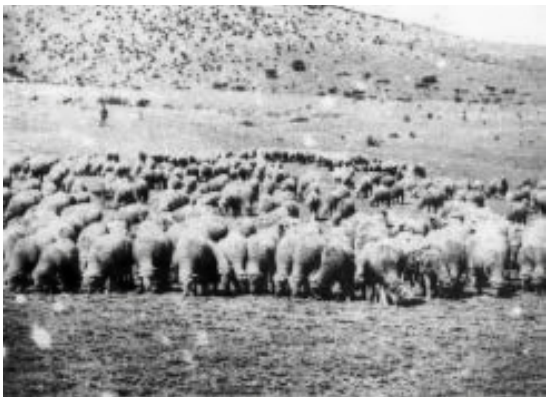
CHACOAN CULTURAL SITE OVERLOOKING THE MIDDLE RIO PUERCO. CABEZON (LEFT) AND CERRO COCHINO (RIGHT) SEEN TO THE NORTH

PRE-20TH CENTURY

Despite inconsistent water, the Rio Puerco watershed has a long history of human inhabitance. Within the watershed, over 10,000 cultural sites have been identified, the oldest of which are dated to 9500 BCE (Curt Larsen and Martha Herzog, 2000). In ancient times the Chacoan people, and later the Ancestral Puebloan people, heavily utilized the region, and their descendants continue to occupy the watershed. In the southern reach of the river, the Pottery Mound Pueblo sat along the banks of the Rio Puerco from 1300-1550 CE, and in modern times is famous for its beautifully painted Kiva murals and finely made ceramics (Franklin, 2018). In more recent history, numerous Indigenous groups have inhabited the region, utilizing the landscape for wild game, medicinal and edible plants, and farming corn, chiles, squash, melons and turkeys (Bryan, 1928; Pueblo of Isleta, 2023; Roger Fragua, personal communication, July 18, 2025). Today, the Acoma, Isleta, Jemez, and Laguna Pueblos, 14 chapters of the Navajo Nation, and the Jicarilla Apache Nation all officially reside within the bounds of the Rio Puerco watershed. Many other Indigenous groups are located nearby.

"Mosquito Man", Kiva Mural Painting from Pottery Mound (Vanpool & Velandia Jagua, 2022).

Prior to the 1800s, the Rio Puerco existed as a small channel with gently sloping sides that frequently flooded the surrounding valley floor (Bryan, 1928). In the 1760s, the Spanish colonial governor approved grazing grants in the watershed, prompting hopeful colonists to move into the region with the intention of grazing livestock. Despite a modest attempt at settlement, Navajo and Apache raiders defending their territory ultimately chased the colonists back to Albuquerque, leaving the valley minimally impacted by livestock (Diggle & Hieb, 2004). In 1868 a treaty was passed in which the Navajo Nation agreed to cease fighting with the United States in return for official rights to their homeland, largely situated outside of the Rio Puerco watershed (Native Knowledge 360, 2019). With the passing of this treaty, colonists were free to bring large herds of livestock into the valley.

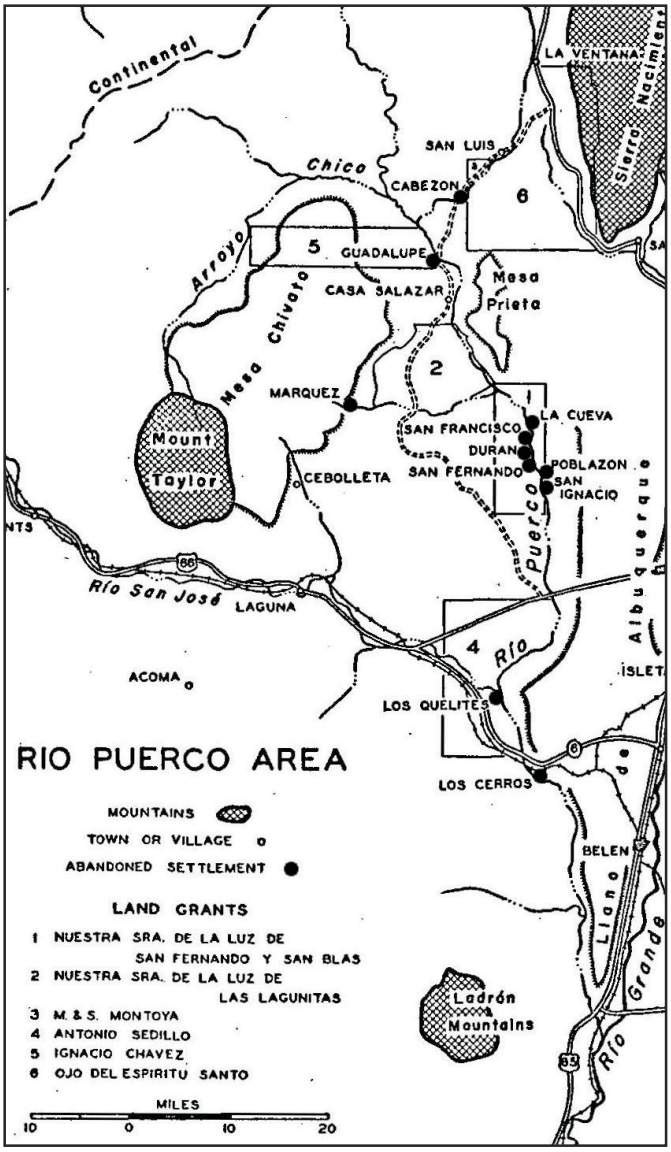


Sheep grazing in the Rio Puerco Valley, 1880s (Rocky Mountain Research Station, 1998).

20TH CENTURY

The valley in the late 1800s was ecologically rich and provided excellent rangeland for livestock. Periodic flooding replenished the land with nutrients and water, and the river was easily dammed and diverted through acequias to irrigate crops. By the end of the 19th century, settlers had thoroughly capitalized on these resources by establishing small towns, building an extensive network of dams and acequias, and grazing vast herds of livestock on both sides of the Rio Puerco (Diggle & Hieb, 2004). News of prosperity in the valley spread, and the Rio Puerco became known as the "breadbasket of New Mexico."

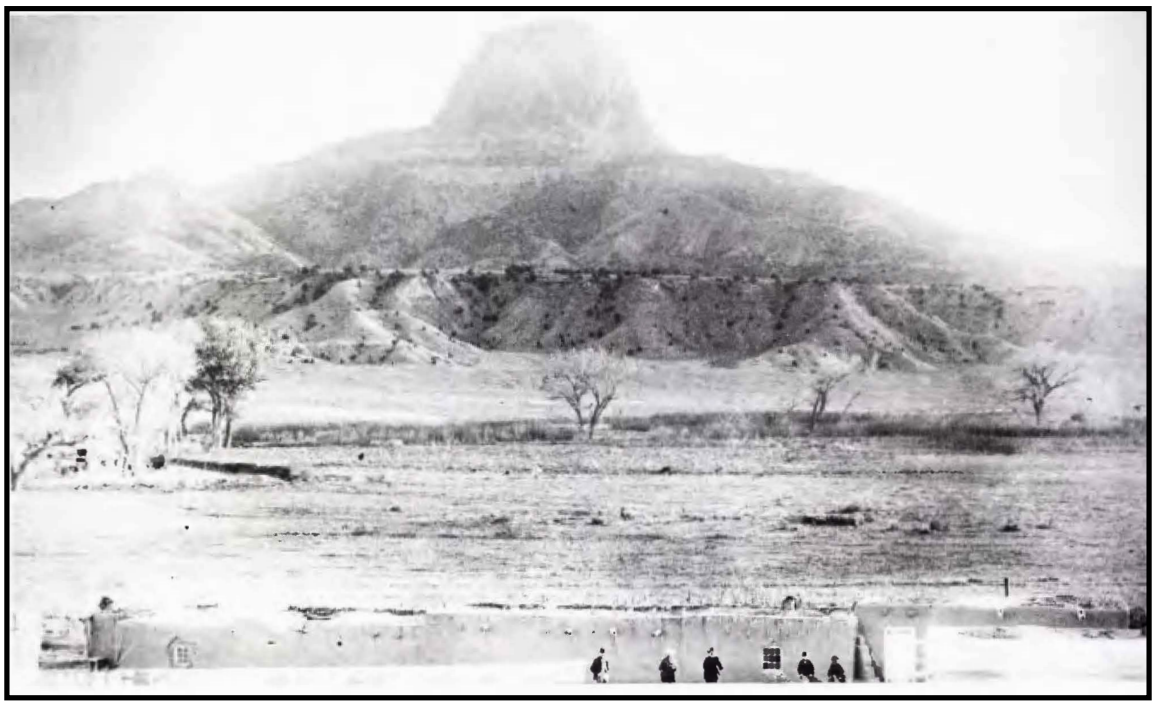
Abandoned structure in the historic town of Guadalupe.



Geography of the Middle Rio Puerco Valley, 1959 (Gwayn Widdison, 1959).

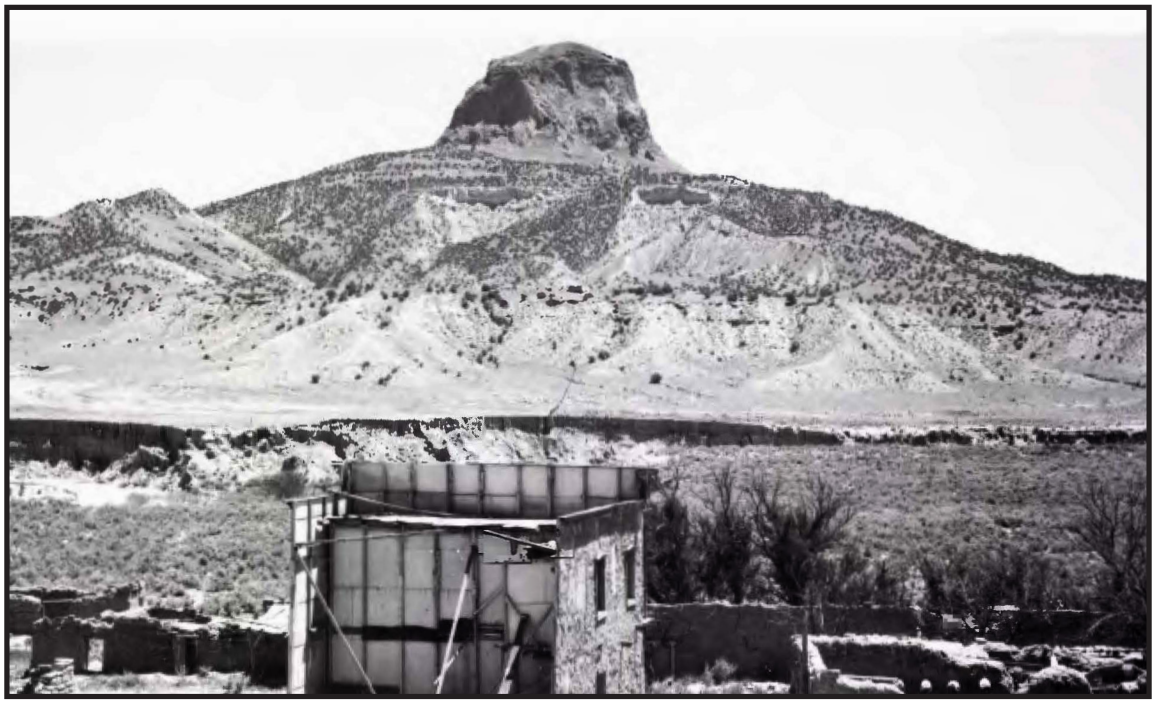
However, this success was relatively short-lived. As early as 1890, the rangelands of the middle valley were reported to be overstocked, and the vegetative cover was quickly eaten away by dense herds of livestock (Gwayn Widdison, 1959). Without the protective barrier of plants, the loose soils of the valley became vulnerable, and accelerated erosion began to dramatically change the shape of the valley. The river channel began to deeply incise, lowering the water table and cutting the river off from its historic floodplain (Diggle & Hieb, 2004). As deep incisions migrated upriver, they dewatered the surrounding valley until what was once a richly vegetated floodplain became sere uplands with little resources to support life. Of the many towns that were established in this brief period of prosperity, many were abandoned, and those that remained became steadily more depopulated and impoverished alongside the land (Bryan, 1928). Nonetheless, the Rio Puerco valley was never abandoned, and today many communities continue to call this watershed home.

1884



The Rio Puerco and Cabezon Peak in the winter of **1884**, looking south from the town of Cabezon. The Rio Puerco flows in a shallow channel marked by cottonwood trees. Cattle and burros graze in the fields. Cottonwood trees indicate a high water table, and the shallow channel is suitable for the practice of flood-irrigation agriculture.

1977



Duplicate photo in **1977**. The Rio Puerco now flows in a deeply incised arroyo 15 meters deep. There has been downcutting of the stream and lowering of the water table, restricting the opportunities for flood irrigation. The fields no longer exist and the town of Cabezon has been abandoned (Lambert & New Mexico Geological Society, 1981).

HISTORICAL PHOTOS

1917

1917 photo looking north of the town of Cabezon (center), with the Rio Puerco below the town. Notice the few cottonwoods lining the banks of the river.

1995

Repeat photograph in **1995**. Notice the massive erosion in front of the town (Rocky Mountain Research Station, 1998).

LANDS
ANAL

THE CONFLUENCE OF THE RIO SAN JUAN WITH THE RIO PUERCO, GOOGLE IMAGERY

SCAPE
LYSIS

GEOGRAPHY | ECOLOGY

The Rio Puerco Watershed is composed of three distinct ecoregions:

Upper Gila Mountains

The Upper Gila Mountains are semi-arid, but contain a variety of microclimates. Higher elevations receive greater amounts of precipitation due to orthographic lift, and streams create wet conditions with dense riparian vegetation. Outside of these microclimates, the landscape is dominated by shrublands and grasslands.

Cold Deserts

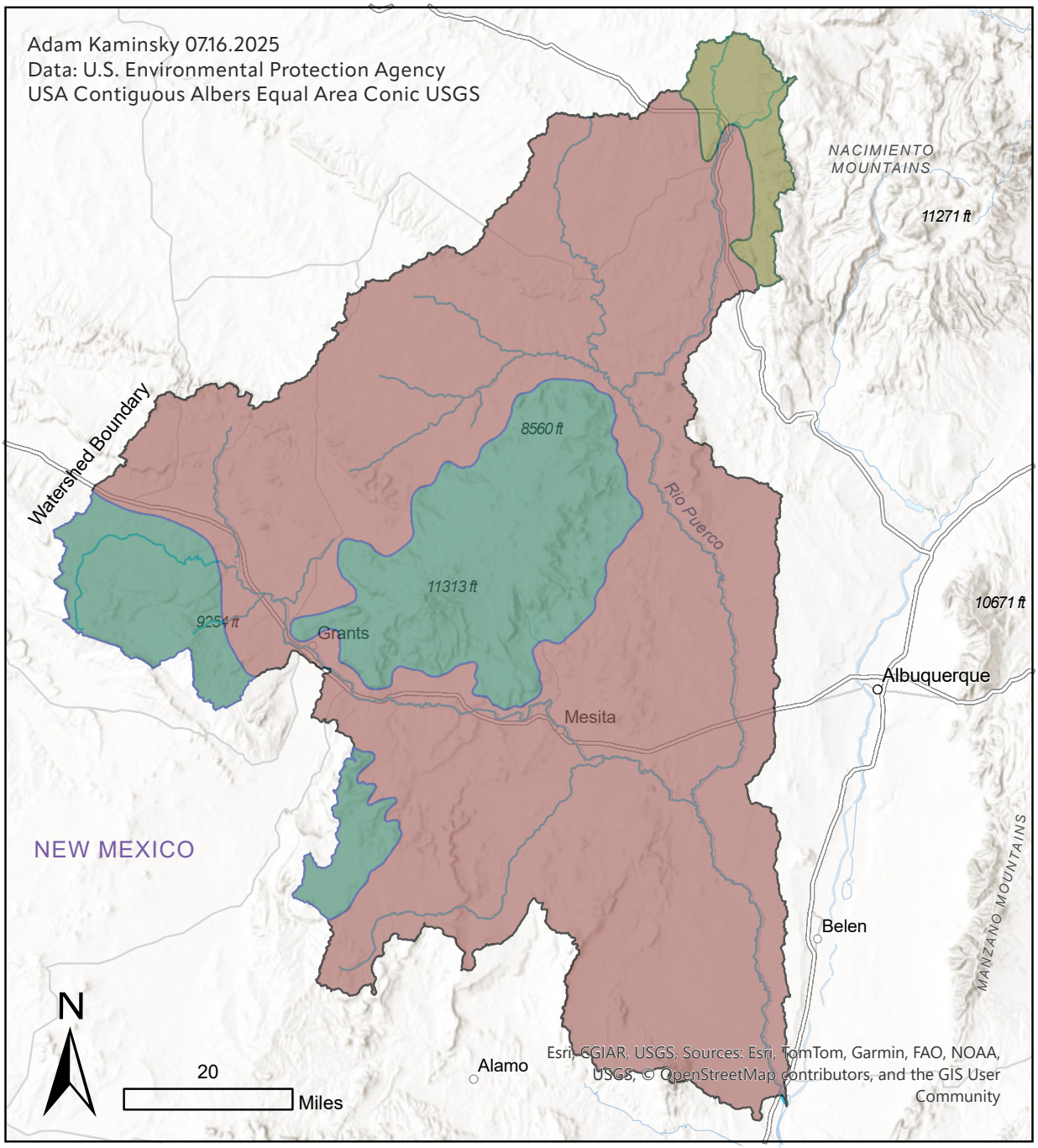
The Cold Desert is characterized by low precipitation and cold winter temperatures. Soils are often alkaline due to retention of calcium in dry conditions, and vegetation consists of sparse shrubs and grasses.

Western Cordillera

The Western Cordillera receives the majority of its precipitation in the form of winter snowfall and summer monsoons, leading to relatively even precipitation throughout the year. Vegetation is dominated by coniferous trees. The topography is often steep, and wildfires burn at regular intervals, leading to a high potential for runoff and erosion.

(Alex Zorach and Jean Epiphan, 2025)

ECOREGIONS OF THE RIO PUERCO WATERSHED



LEVEL II ECOREGIONS

UPPER GILA MOUNTAINS

COLD DESERTS

WESTERN CORDILLERA

SOIL

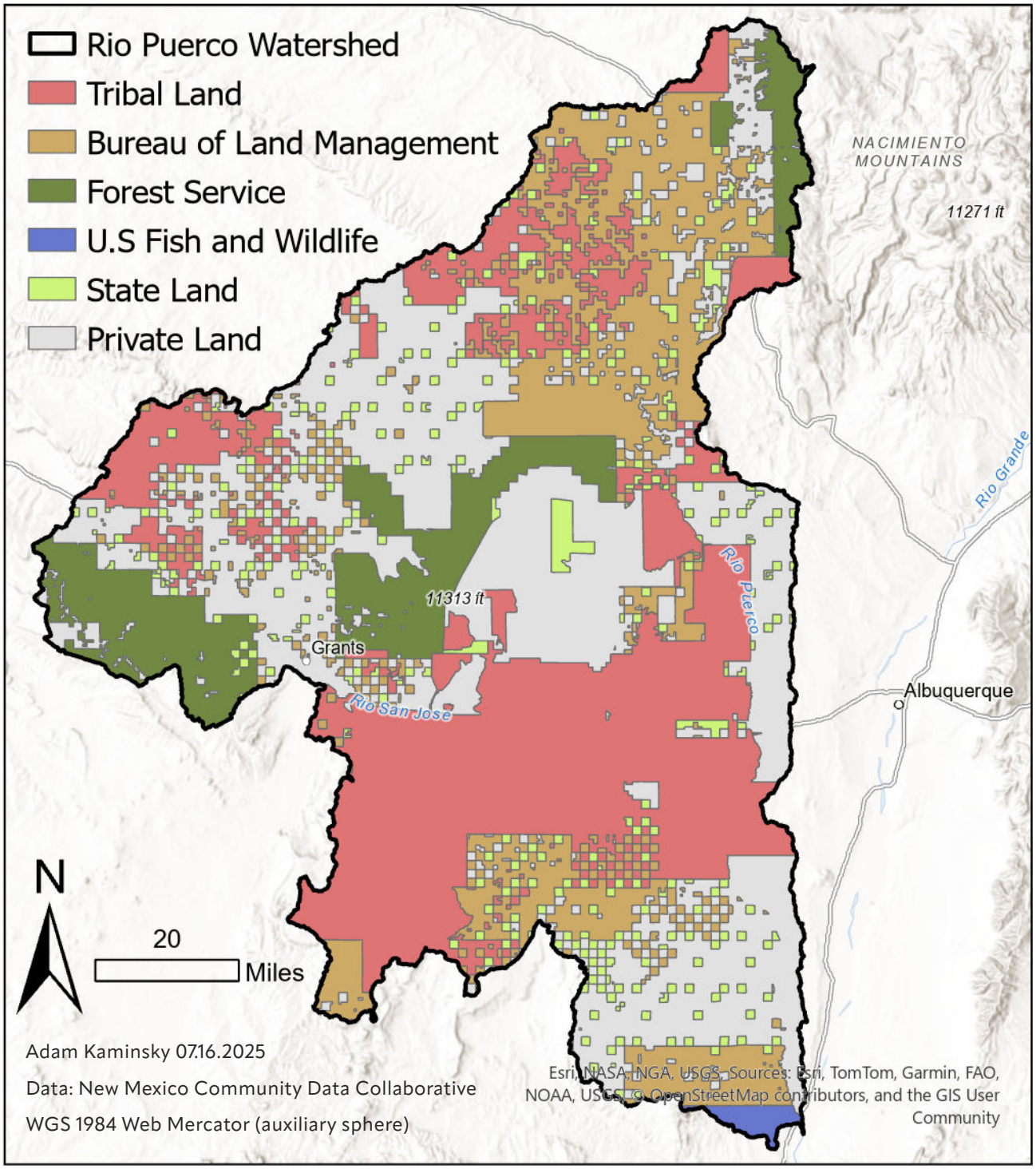
Numerous studies indicate that typical soils within the watershed are **highly erodible** and have **low infiltration** rates, leading to **extreme erosion and runoff**.

(Office of the State Engineer, 2020; Phippen & Wohl, 2003; Raymond D. Watts et al., 2000)

However, the long history of land use in the watershed indicates that these soils have significant **ecological potential** if erosion is carefully managed.

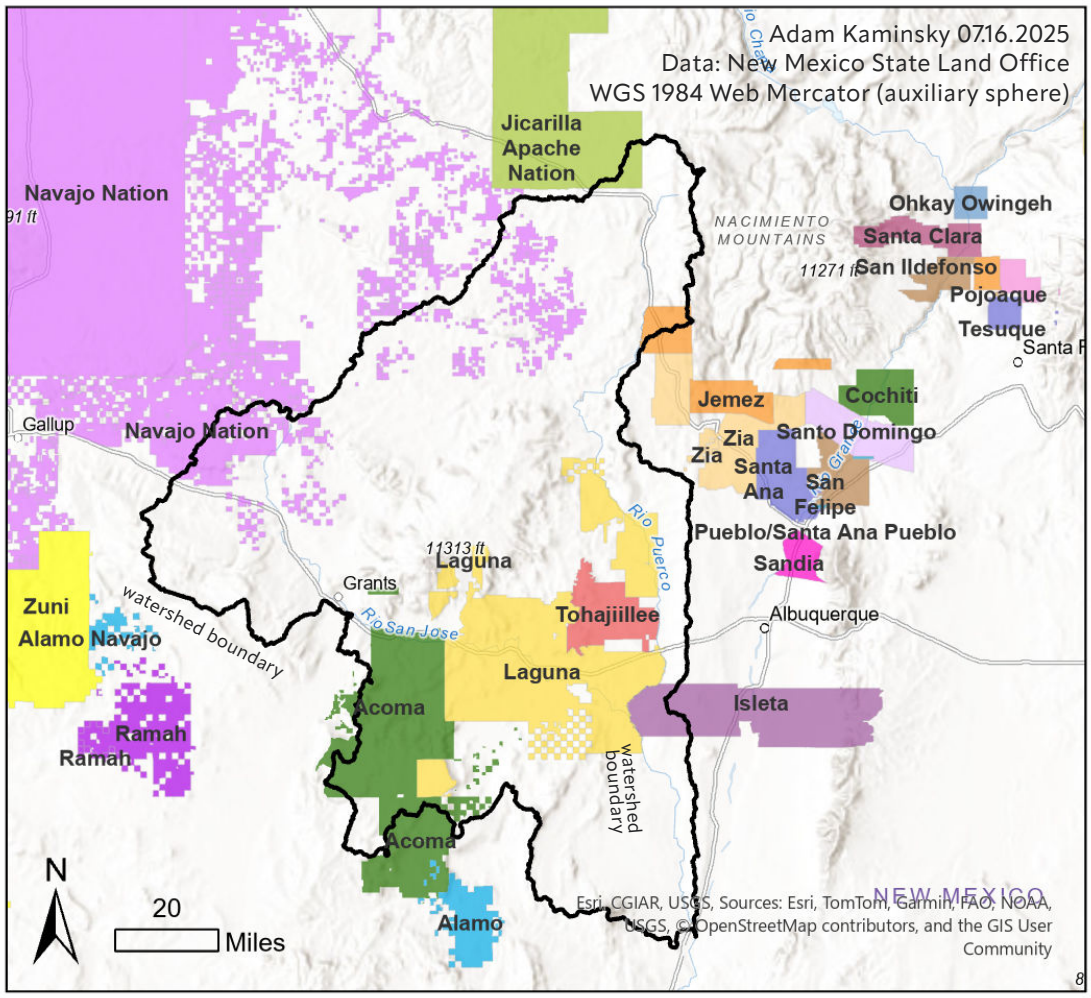
LOOKING SOUTHWEST ACROSS ARROYO CHICO WITH MESA CHIVATO IN THE BACKGROUND

LAND OWNERSHIP

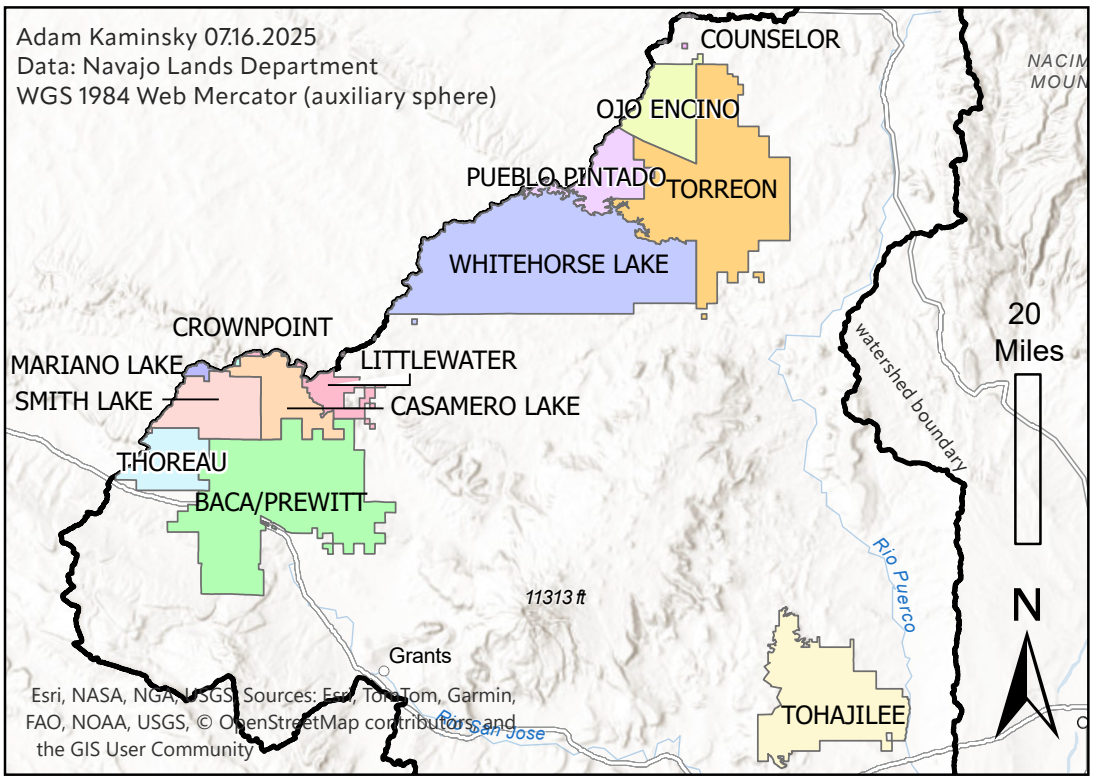


LEGAL LAND OWNERSHIP IN THE RIO PUERCO WATERSHED

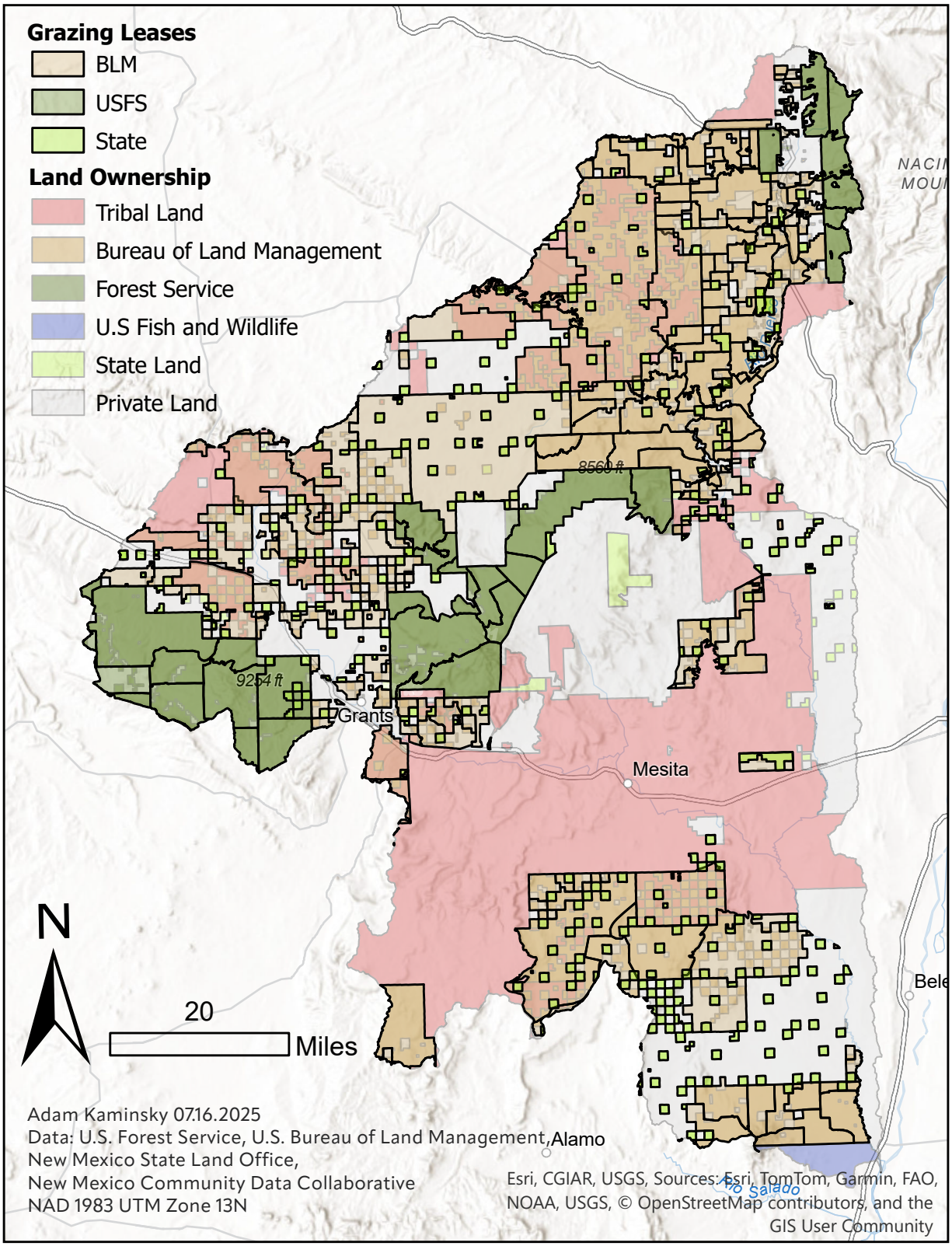
TRIBAL LANDS



NAVAJO CHAPTERS



GRAZING LEASES



GRAZING LEASES IN THE RIO PUERCO WATERSHED

Although land ownership in the watershed is diverse, large swaths are owned and managed by a few entities, principally **Tribal Governments** and the **BLM**.

Onion Blanketflower in the Lower Rio Puerco.

Collaboration with these entities presents an opportunity to connect with a broad community of people who are **economically**, **socially**, and **spiritually** connected to the land.

RANGELAND POTENTIAL

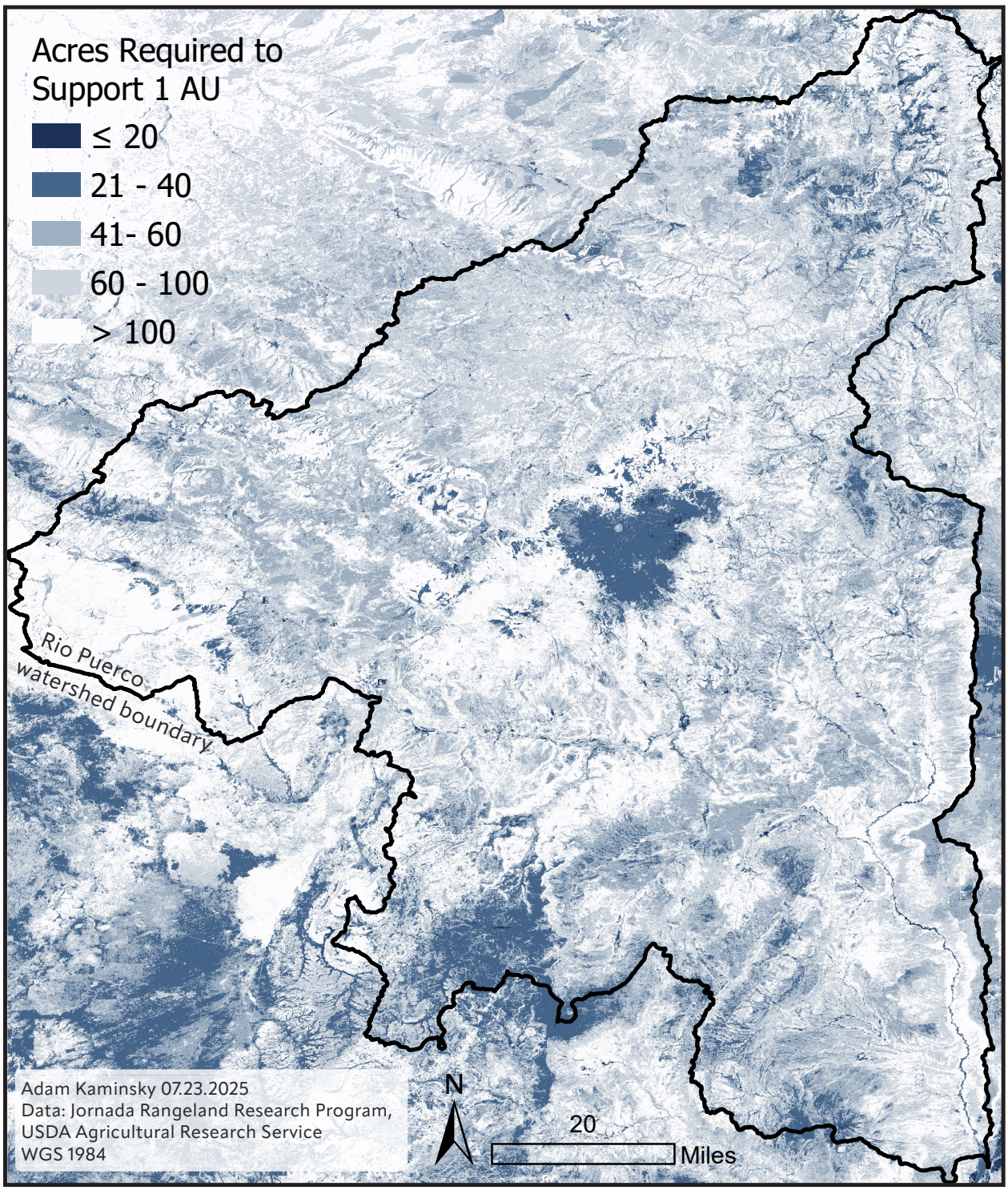
The Rio Puerco watershed has been overgrazed in the past, and can now appear too barren to sustain livestock. However, according to annual biomass growth data, many areas within the watershed grow sufficient grasses and forbs to support cattle, meaning that sustainable ranch management could be a viable restoration strategy within the watershed.

Looking south across the Middle Rio Puerco, two calves escaped their fence.

These data were calculated based on average cattle grazing needs at a 50% utilization rate for 360 days. Data visualization breaks were based on the literature and conversations with experts at NRCS and NMSU Agricultural Extension. Biomass data from the Rangeland Analysis Platform were summed to represent both annual and perennial forbs and shrubs, and then averaged across the past five years. This analysis does not include complicating factors that can dramatically change grazing potential, and should therefore be used only as a general approximation (Beef Cattle Research Council, n.d.; Casey Spackman, personal communication, July 18, 2025; Casey Spackman & Marcy Ward, 2015; DelCurto et al., 2023; Jim Armendariz, personal communication, July 23, 2025; Lee Rinehart, 2020; University of Nebraska Lincoln, n.d.).

On average, ranchers in this region of New Mexico need 40-60 acres to support one animal unit (AU). On this map, lower acreage per AU indicates higher rangeland potential.

RANGELAND POTENTIAL BASED ON ANNUAL FORAGE BIOMASS



WATE
THR

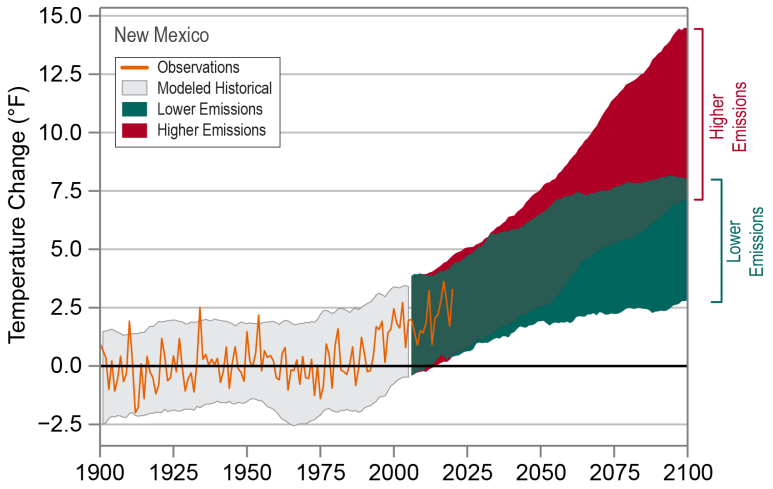
RSHED
EATS

HEAVILY ERODED FENCELINE IN THE MIDDLE RIO PUERCO WITH CERRO COCHINO AND PRIETA MESA SEEN TO THE EAST

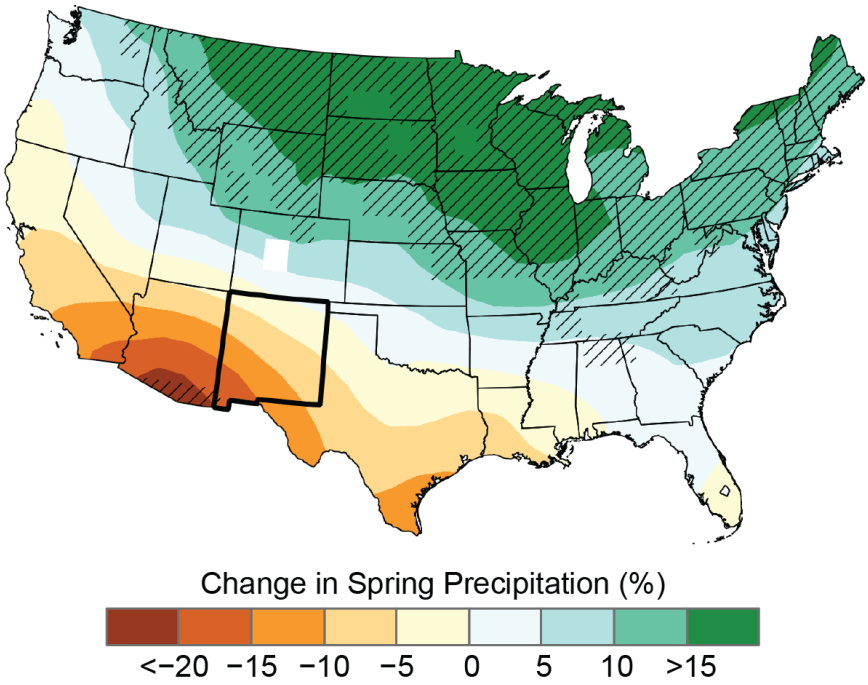
CLIMATE

Climate change has already begun to affect the Rio Puerco region, and is predicted to have increasingly significant effects.

New Mexico is anticipated to continue experiencing hotter and drier weather, with extreme conditions concentrated in the spring. Traditionally, snowmelt in the spring sustains New Mexico rivers until the monsoon season begins in the summer, so the new weather patterns are predicted to significantly worsen drought and increase the ephemerality of New Mexico river systems.



Projected temperature change in New Mexico (Rebekah Frankson & Kenneth Kunnel, 2022).



Projected changes in total spring (March–May) precipitation (%) for the middle of the 21st century compared to the late 20th century under a higher-emissions pathway (Rebekah Frankson & Kenneth Kunnel, 2022).

Furthermore, the dependable monsoonal pattern — which annually contributes the majority of precipitation in the Rio Puerco — is expected to become more erratic, leading to highly variable drought conditions in the future (Molnár & Ramírez, 2001; Rebekah Frankson & Kenneth Kunnel, 2022).

WATER QUALITY

The Rio Puerco is considered one of the most highly degraded rivers in the United States.

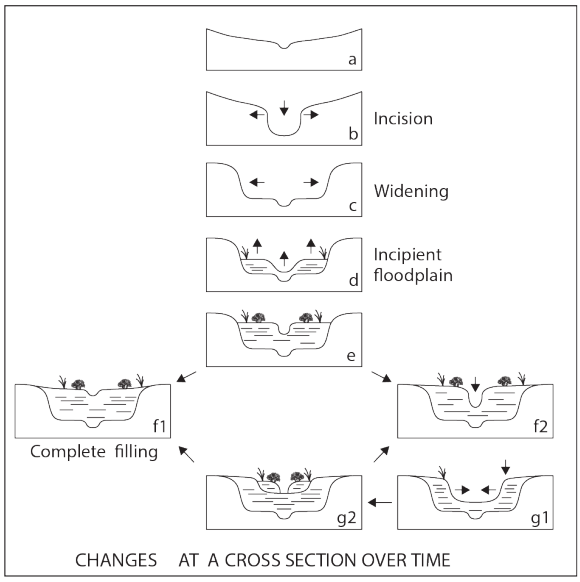
In 2010 the New Mexico Environment Department completed a comprehensive study of water quality in the Rio Puerco and found **sediment**, **temperature**, and **nutrients** to be the major polluting factors (Hopkins & Coleman, 2010).

Tributary to the Middle Rio Puerco near Arroyo Chico.
Looking southwest with Mesa Chivato in the background.

WATER QUALITY | SEDIMENT

Although the underlying geology in the Rio Puerco leads to naturally high sediment loads, the watershed contains higher sediment than would be expected through natural processes alone (Hopkins & Coleman, 2010).

Famously, although the Rio Puerco contributes only 10% of the total water flow to the Rio Grande, it contributes up to 80% of the total sediment load (Meg Hennessey, 2017). Overgrazing is often cited as a major cause of erosion and sediment in the river, and numerous studies have found significant links between grazing intensity and sediment yields (Gellis et al., 2001, 2012, 2017; Meg Hennessey, 2017; Phippen & Wohl, 2003). Overgrazing decreases plant biomass, which increases bare ground that is susceptible to erosion. Furthermore, overgrazing in riparian areas reduces plant communities along riverbanks, destabilizing banks and leading to increased erosion.



Stages of arroyo evolution at one place over time (Hopkinds and Coleman,2010)

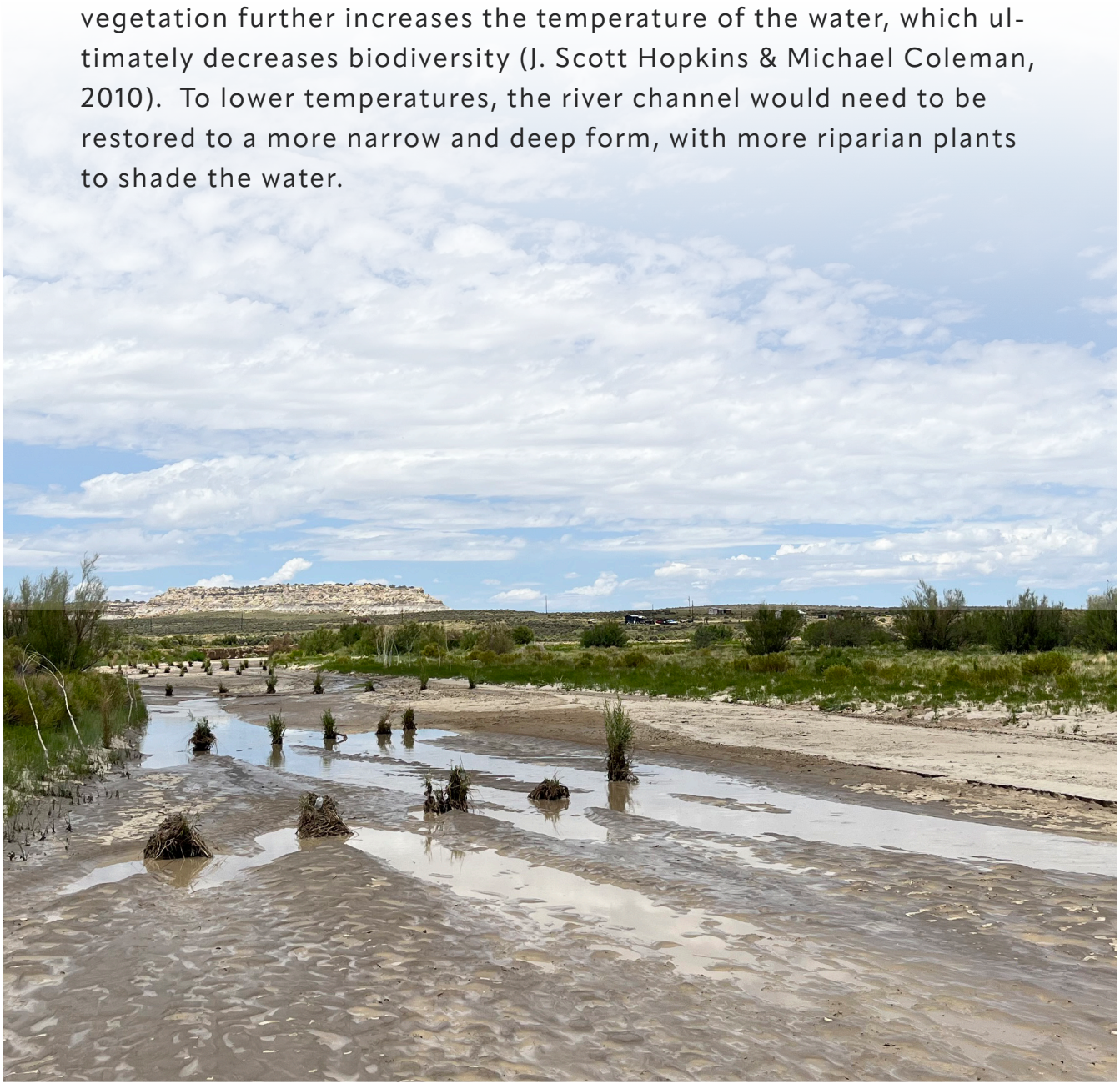
Although overgrazing has been linked to erosion, some studies assert that the cause of erosion in the Rio Puerco is more nuanced. Geological studies of the riverbed indicate that the Rio Puerco has undergone periodic and intense cycles of erosion and aggradation (filling in), predating the erosion due to grazing in the 20th century (Gellis et al., 2017). Furthermore, unpaved roads have been found to contribute significantly to erosion in the Rio Puerco watershed (Phippen & Wohl, 2003). The watershed is full of unpaved roads in varying conditions, so targeting unused roads could have significant potential for restoration.

Despite high sediment loads and erosion, the Rio Puerco is in a period of aggradation, meaning that the majority of the sediment carried by the river is being deposited in the channel. Arroyos within the Rio Puerco are filling and narrowing, and in some cases the entire channel is predicted to fill within 150 years (Gellis et al., 2017). Natural aggradation will eventually reverse the damage of extreme erosion, but restoration efforts can accelerate this process in order to aid the communities that currently rely on the Rio Puerco watershed.

WATER QUALITY | TEMPERATURE

Temperature is another major form of pollution in the Rio Puerco.

Within the streambed itself, erosion leads to shallow, wide, and straight channels, which increases the surface area for sunlight to penetrate the water. A reduction in shade from degraded riparian vegetation further increases the temperature of the water, which ultimately decreases biodiversity (J. Scott Hopkins & Michael Coleman, 2010). To lower temperatures, the river channel would need to be restored to a more narrow and deep form, with more riparian plants to shade the water.

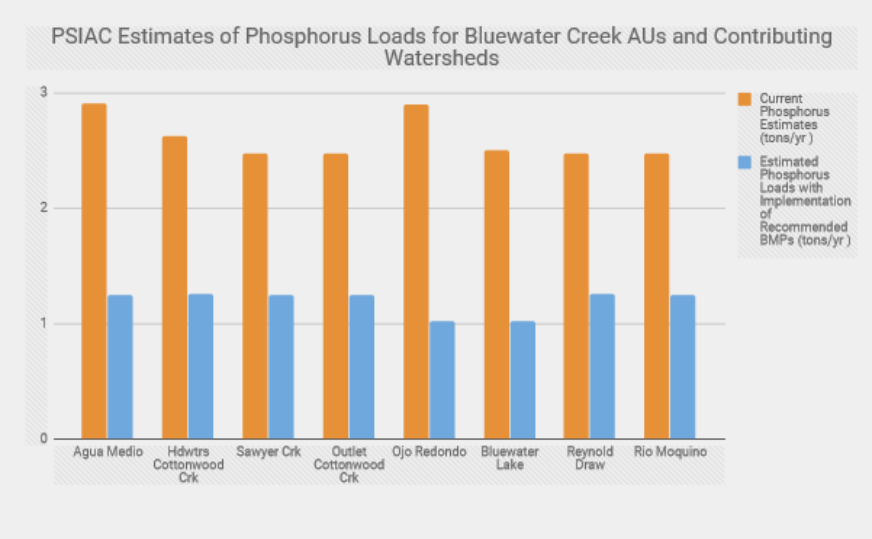


Torreon Wash with Willow plantings to catch sediment and restore stream architecture.

WATER QUALITY | OTHERS

Nutrients:

Certain tributaries of the Rio Puerco have excessive nutrient loads, which is detrimental to water quality due to increased plant growth and die-offs that can dramatically reduce dissolved oxygen. The majority of nutrient pollution has been attributed to runoff from agricultural land, while some point-source pollution has been identified originating from the Cuba Wastewater Treatment Plant. Increasing the health of rangelands would reduce runoff of nutrients, and increasing riparian vegetation would buffer the waterways from the effects of runoff (J. Scott Hopkins & Michael Coleman, 2010).



Phosphorus loads in the Bluewater Creek, a tributary to the Rio Puerco. Measured loads are shown in Orange, while Blue depicts predicted loads under the recommended Best Management Practices (BMPs) (Meg Hennesey, 2017).

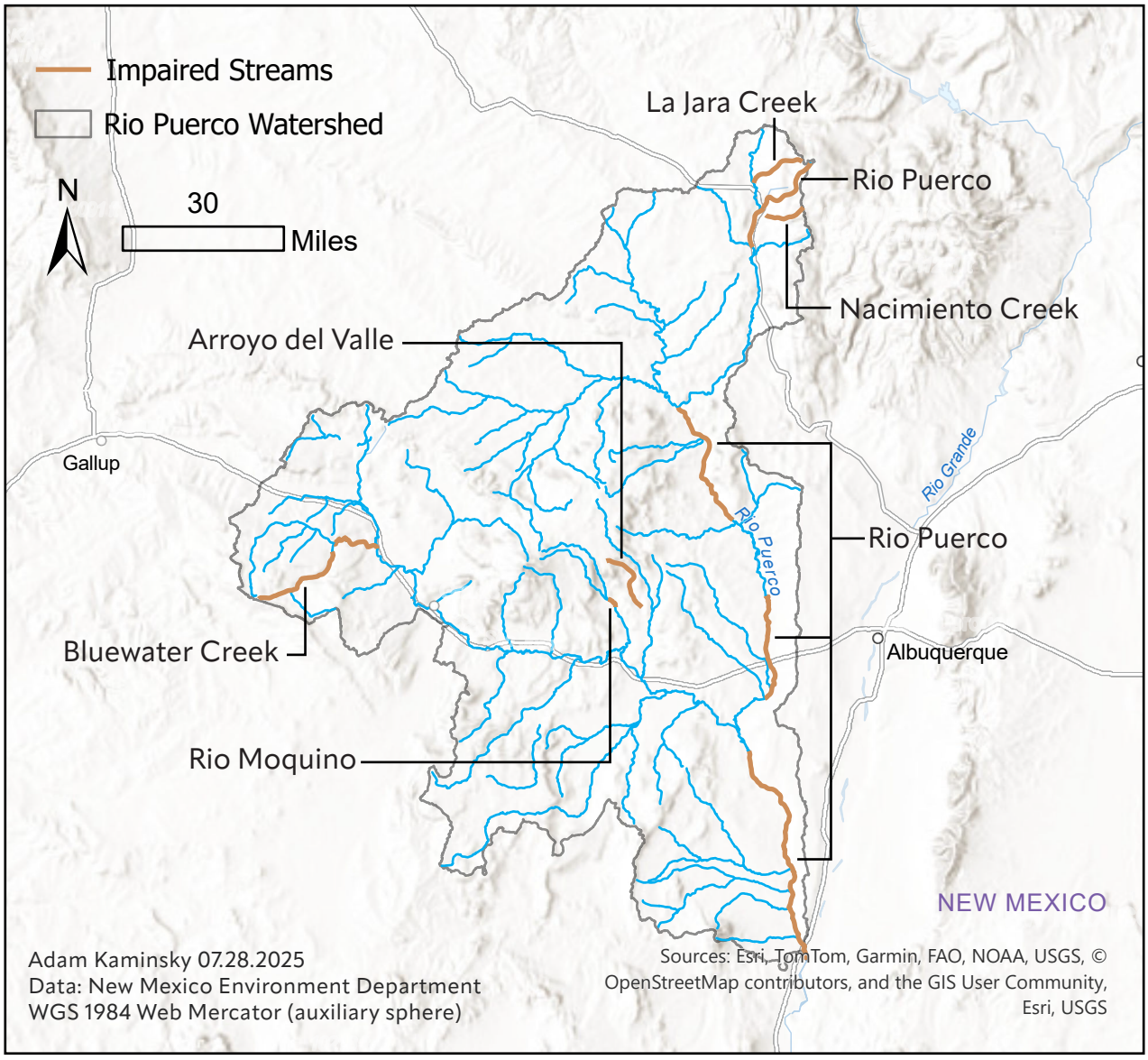
Aluminum:

Elevated aluminum contamination has been found in the Rio Puerco watershed, but always at levels below toxicity for fish. There are no anthropogenic sources of aluminum in the watershed, so the cause has been attributed to acidic snowmelt leeching aluminum from the soil (J. Scott Hopkins & Michael Coleman, 2010).

Snowmelt example from the Northern Rockies.

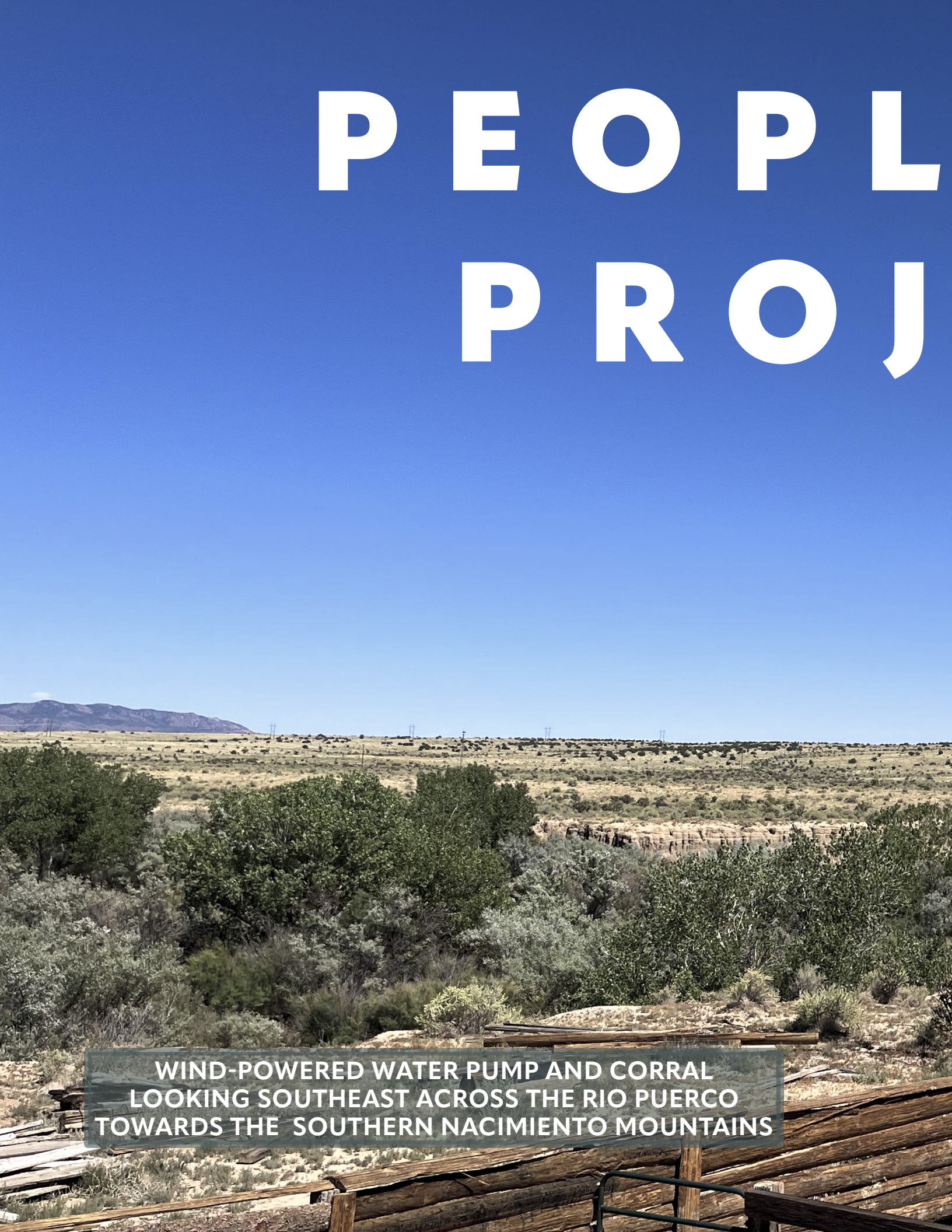
IMPAIRED WATERWAYS

Although the entire Rio Puerco suffers from sedimentation and temperature pollution, certain segments are officially designated as impaired by the New Mexico Environment Department, serving as potential areas of focus for restoration efforts.



IMPAIRED STREAMS IN THE RIO PUERCO WATERSHED

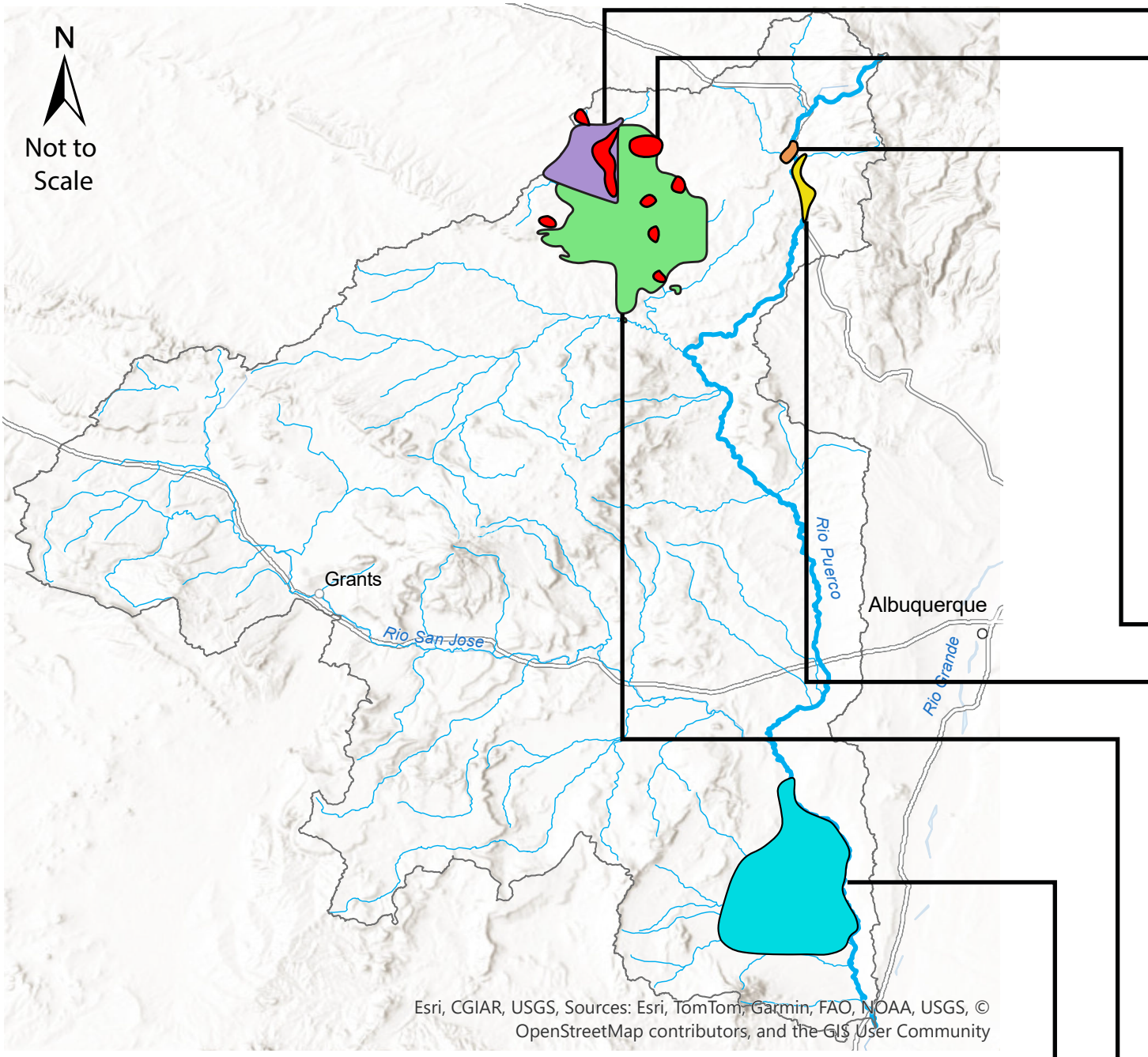
PEOPL
PROJ



WIND-POWERED WATER PUMP AND CORRAL
LOOKING SOUTHEAST ACROSS THE RIO PUERCO
TOWARDS THE SOUTHERN NACIMIENTO MOUNTAINS

E AND ECTS

PAST AND CURRENT PROJECTS



Comanche Ranch
Pueblo of isleta

The Comanche Ranch property was purchased by the Isleta Pueblo in 2016. They ran cattle for many years, but have since stopped due to budgetary and political complications. Since 2016, the ranch has installed dozens of erosion control structures, and experimented with traditional and regenerative grazing practices. In 2023 they submitted a grant to restore large areas of land, particularly around the Pottery Mound site.
isletapueblo.com/home/environment-department/

The Ojo Encino chapter of the Navajo Nation has a long history of improving rangeland through traditional grazing practices, earning them the Quivira Coalition's Clarence Burch Award in 2007. Currently, members of their community collaborate with River Source to install and monitor erosion control structures.
ojoencino.navajochapters.org

In the past, River Source collaborated with the Rio Puerco Alliance to install and monitor erosion control structures. Rio Puerco Alliance is less active now, but River Source has many active projects in the Torreon Wash.
riversource.net

Rio Grande Return has worked to control erosion in the main stem of the Rio Puerco. They recently assumed stewardship over two grazing leases on State Land along the Rio Puerco, and plan to practice restoration on these sites.
riograndereturn.org

The BLM has been working in the Rio Puerco Watershed for decades. They operate the Rio Puerco Management Committee, and have many active projects south of Cuba and throughout the watershed. Importantly, the BLM also controls vast areas of land through grazing leases, which could play a critical role in traditional/regenerative grazing in the future.
blm.gov/get-involved/resource-advisory-council/near-you/new-mexico/rio-puerco-management-committee

The Torreon Chapter of the Navajo Nation has a sizable parcel of land in the Rio Puerco watershed. It is unclear what grazing practices they employ, but they have collaborated with non-profits in the past to implement traditional/regenerative grazing and erosion control. They may be an important partner for future grazing restoration work.
torreon.navajochapters.org

Jicarilla Apache Nation

The Jicarilla Apache Nation is largely outside of the Rio Puerco watershed. However, they have excellent examples of traditional/ regenerative ranching elsewhere on their land, which could be brought into the watershed.
facebook.com/jicarillaapachenationpr

Pueblo of Jemez

The Jemez Pueblo has an organized rangeland management program, and members of the Pueblo actively practice traditional/ regenerative ranching.
jemezpueblo.org/government/natural-resources

Eastern Chapters Navajo Nation

The Torreon and Ojo Encino chapters of the Navajo Nation already have connections to Rio Puerco restoration. Many other chapters might be active in watershed restoration as well, but their participation is unknown. More information is needed to understand the role of these chapters in the watershed.

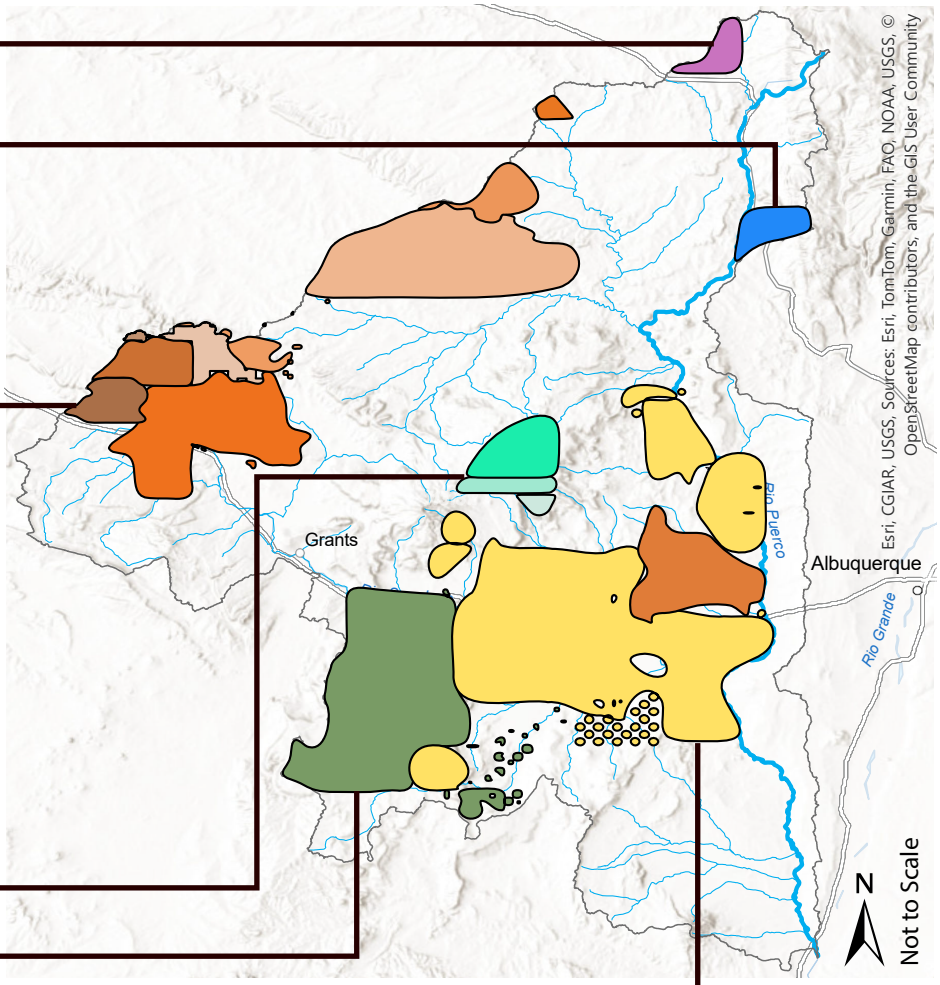
Private Landowners

The plateau north of Mt. Taylor exhibits high grass and forb growth, giving it great potential for regenerative grazing. Much of this land is owned by 3 private corporate landowners: Fernandez Wind Energy (North), Fernandez Co Limited (Middle), and Lobo Partners (South). These companies might be interested in collaborating on Rio Puerco restoration.

Acoma Pueblo

The Acoma Pueblo manages a large tract of land in the Rio Puerco watershed, much of which exhibits high rangeland potential based on annual biomass data. Their Natural Resources Department manages Pueblo Rangeland.
puebloofacoma.org

FUTURE COLLABORATIONS



Laguna Pueblo

The Laguna Pueblo controls the largest tract of continuous Pueblo land in the Rio Puerco Watershed, which contains a long section adjacent to the Rio Puerco itself. Their Rangeland Management Program or Natural Resources Department might be interested in collaborating on Rio Puerco restoration.
lagunapueblo-nsn.gov/departments/environmental-and-natural-resources/range-management-program

Science

Increased scientific engagement through universities and independent organizations could help practitioners and land stewards better understand and manage the watershed. Specifically, a growing body of research, led by an organization called CrustNet, demonstrates the efficacy of biological soil crusts as tools for restoration. Given the aridity and abundance of bare soil in the watershed, biocrusts could play a critical role in restoration, and CrustNet could help provide resources and expertise in this field.

CONCLU

USIONS

LOOKING WEST ACROSS THE UPPER RIO PUERCO FROM
BLUE BIRD MESA IN THE NACIMIENTO MOUNTAINS

RECOMMENDATIONS

This report identifies the need for collaborative restoration in the Rio Puerco watershed, and recommends the following interventions:

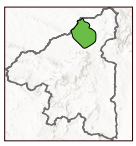
1: Improve habitat and control erosion within the Rio Puerco and surrounding upland ecosystems.

To achieve this goal, people and organizations will need to actively pursue restoration projects in the watershed. A substantial body of literature and a rich traditional ecological tradition describe numerous methods of controlling erosion and restoring upland, riparian, and aquatic ecosystems. However, time and resources are common roadblocks to enacting these methodologies.

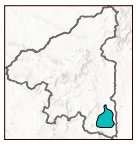
This report provides information and data supporting the need for restoration, and the findings of this report may be included in grants to potential project funders. The report also indicates areas of focus for future work, and identifies the major organizations with connections to the watershed.

2: Foster a community of practice that communicates regionally in pursuit of innovative land management and erosion control strategies.

To create a community of practice around restoration, it is first necessary to identify potential community members. This report identifies many people and organizations with connections to the land, and therefore a potential stake in restoration efforts. Two communities stand out as potential starting points for a broader community of practice:



The **Ojo Encino Ranching Committee** and **Ranchers in the Torreon Chapter of the Navajo Nation** are currently involved in restoring the watershed in collaboration with **River Source** and **Rio Grande Return**.



The **Comanche Ranch** in the **Isleta Pueblo** has a history of rangeland restoration and erosion control, and may be interested in rejuvenating restoration efforts.

From these starting places, a broader community of practice could be established that incorporates the people and organizations listed on the "Future Collaborations" page, with a regional focus on sustainable grazing practices in areas with high rangeland potential.

3: Increase collaboration among relevant organizations across the fields of agriculture, conservation, and restoration.

Collaborative watershed restoration is enriched by expanding to a broad community of farmers, ranchers, scientists, planners, and practitioners. This report identifies that the **BLM, Rio Grande Return, and River Source (with a longtime independent contractor named Watson Castillo)** are currently contributing to restoration in the watershed at this time. However, it is likely that numerous Tribal Governments are actively working towards conservation and restoration as well.

Restoration planners and practitioners currently dominate the watershed restoration effort, so this report recommends diversifying collaboration to include farmers, ranchers, and scientists via **Tribal Governments**, **ranchers utilizing BLM leases**, **private land owners**, **CrustNet** and **New Mexico colleges and universities**.

Cerro de Guadalupe at sunrise with the Rio Puerco just behind.
Wildfire smoke streaming over Mesa Chivato in the background.

CONCLUSION

This report demonstrates a need for restoration in the Rio Puerco watershed, and maps the community of people and organizations working on the land. The intention of this report is to inform and support the people and organizations dedicated to improving the ecological, economic, and social health of the Rio Puerco watershed, and may be widely used, distributed, and built upon to meet that goal.

For questions, comments, or to request map or image files, please contact:

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ACKNOWLEDGMENTS

This document was created with the help of many generous individuals. Special thanks to Elicia Whittlesey (Quivira Coalition), who made this project happen and provided essential guidance throughout the process. Technical support from Kristina Young (The Jornada Rangeland Research Program), Nicole Scholtz (University of Michigan), Jim Armendariz (University of New Mexico Ag. Extension), and Max Hewes (Boise State University). New Mexican cultural understanding from the many people at the Grazing Together Conference, Warren Edaaki (NM Healthy Soils Working Group), and Roger Fragua (Wildflower Institute). Design guidance from Amanda Szot (University of Michigan) and Gary Oakley (artist). General guidance from Sarah Wentzel-Fisher (Thornburg Institute), Reid Whittlesey (Rio Grande Return), Rich Schrader (River Source), Emily Wolf (American Rivers), Toner Mitchell (Trout Unlimited), Jan-Willem Jansens (Ecotone Landscape Planning), and Ellie Oakley (The Nature Conservancy).

Front Cover: Lower Rio Puerco near its confluence with the Rio Grande seen from an airplane with Ladron Peak in the Background.
Back Cover: Lower Rio Puerco with Ladron Peak in the background

Looking southeast at an eroded tributary to the Middle Rio Puerco near the confluence with Arroyo Chico. Numerous volcanic "Cerros" and Prieta Mesa (back left) seen in the background.

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Adam grew up amidst forests, swamps, and farms in Vermont, and earned a degree in Conservation Biology from Middlebury College. He spent his twenties in Montana, where he worked in outdoor education, organic farming, and ecological restoration, and spent five years as the Education Director on an organic farm. He is now pursuing a degree in Landscape Architecture and Ecosystem Science and Management at the University of Michigan, and hopes to continue working in rural landscape restoration and community development.

Adam and a juniper in front of Cabezon in the Middle Rio Puerco.